Confidence Intervals & Hypothesis Testing

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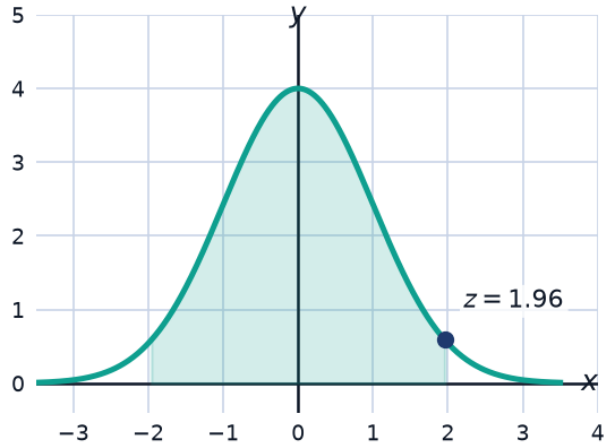
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Constructing a confidence interval

A random sample of 50 students has a mean study time of $\bar{x} = 6.2$ hours per week, with sample standard deviation $s = 1.8$ hours. Construct a 95% confidence interval for the population mean study time, using a critical value of $z^* \approx 1.96$.

2 [Foundational] • Target: Constructing a confidence interval

Interpret the confidence interval from the previous problem in the context of the study time question, using correct statistical language.



3 [Foundational] • Target: Margin of error and sample size

Explain two ways to decrease the margin of error of a confidence interval, and state a tradeoff involved with each.

4 [Foundational] • Target: The logic of hypothesis testing

State the null hypothesis H_0 and alternative hypothesis H_a for a test of whether a coin is biased toward heads (true proportion of heads p), and classify the test as one-tailed or two-tailed.

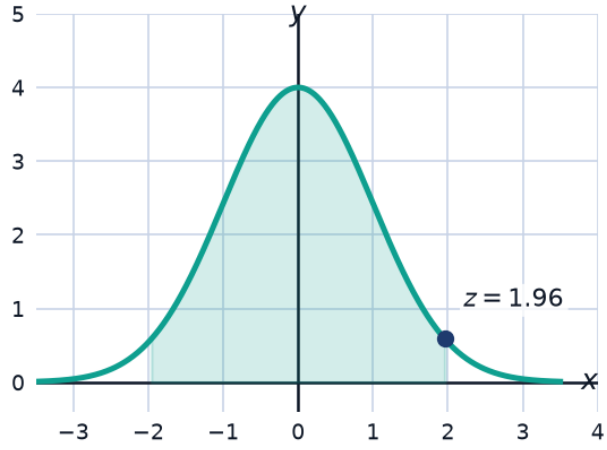
5 [Foundational] • Target: The logic of hypothesis testing

Explain what a p-value represents, in the context of a hypothesis test.

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6 [Foundational] • Target: Four-step hypothesis testing procedure

A manufacturer claims their light bulbs last on average $\mu = 1000$ hours. A sample of 40 bulbs has mean lifetime $\bar{x} = 985$ hours with sample standard deviation $s = 45$ hours. State the hypotheses and compute the test statistic for testing whether the true mean lifetime is less than claimed.



7 [Foundational] • Target: Four-step hypothesis testing procedure

Using the test statistic $z \approx -2.11$ from the previous problem and a p-value of approximately 0.017, state a conclusion at the $\alpha = 0.05$ significance level.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Type I and Type II errors, and power

In the light bulb test above, describe what a Type I error would mean in context, and what a Type II error would mean in context.

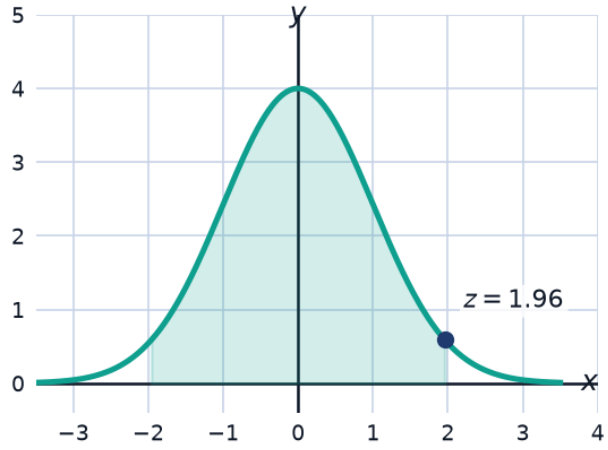
9 [Intermediate] • Target: Type I and Type II errors, and power

Explain what the power of a test measures, and state one way to increase it.

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10 [Intermediate] • Target: Confidence intervals for a proportion

A poll of 400 randomly selected voters finds that 220 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.



11 [Intermediate] • Target: Determining sample size for a target margin of error

A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a conservative planning value of $p = 0.5$ (which maximizes required sample size). Find the minimum sample size needed, using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with $z^* = 1.96$.

12 [Intermediate] • Target: One-sample t-test for a mean

A nutritionist claims a new granola bar has a mean of 200 calories. A sample of 15 bars has mean $\bar{x} = 206$ calories with sample standard deviation $s = 9$ calories. Since the population standard deviation is unknown and n is small, a t-test is used. Compute the test statistic $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$.

13 [Intermediate] • Target: One-sample t-test for a mean

Explain what it means for a hypothesis test result to be "statistically significant," and why statistical significance does not always imply practical importance.

14 [Intermediate] • Target: One-sample t-test for a mean

A researcher wants to test whether a new fitness program changes the average resting heart rate of participants from the population baseline of $\mu = 72$ bpm. A sample of 36 participants after the program has mean resting heart rate $\bar{x} = 68.5$ bpm with sample standard deviation $s = 10.5$ bpm. State the hypotheses (choosing a two-sided alternative), compute the test statistic, and using a p-value of approximately 0.11, state a conclusion at $\alpha = 0.05$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

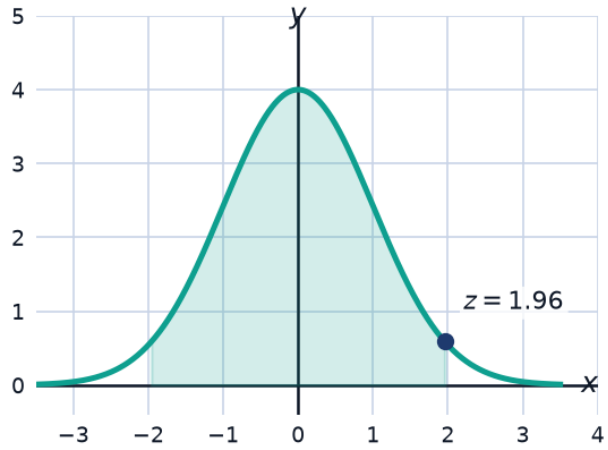
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15 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 2.01 Sampling Methods & Bias

Explain the relationship between a confidence interval and a two-sided hypothesis test at the same confidence/significance level — specifically, describe how you could use a 95% confidence interval alone to decide whether to reject $H_0 : \mu = \mu_0$ at $\alpha = 0.05$, without separately computing a p-value.

16 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A city council is deciding whether to fund a new public transit line, and wants to survey residents to estimate the true proportion who would use it regularly. A preliminary small survey suggests around 35% support, but the council wants a formal survey with a margin of error no larger than 2.5% at 95% confidence. Using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with the preliminary estimate $p = 0.35$ and $z^* = 1.96$, find the required sample size, rounding appropriately.



17 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A restaurant chain wants to test whether a new menu item's average preparation time is longer than the target of 8 minutes, since customers have complained about slow service. A random sample of 30 orders for the new item has a mean preparation time of $\bar{x} = 8.7$ minutes with sample standard deviation $s = 1.8$ minutes. State the hypotheses for this one-sided test, compute the test statistic, and using a p-value of approximately 0.019, state a conclusion at $\alpha = 0.05$ in the context of the restaurant's concern.

18 [Challenging] • Target: Confidence intervals for a proportion • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

An agricultural researcher is designing an experiment to study the yield of a new variety of wheat. The experiment investigates two factors: - Factor 1: Fertilizer formula, with 3 different types. - Factor 2: Watering frequency, with 4 different schedules. If the experiment uses a completely randomized design where each subject receives a combination of one fertilizer formula and one watering schedule, how many distinct treatment groups are there?

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19 [Challenging] • Target: Determining sample size for a target margin of error • Cross-Lesson: 3.03 The Normal Distribution

A study investigates the effectiveness of a new teaching method by comparing the test scores of students who used the new method versus those who used the traditional method. The results are summarized in a two-way table. If the table has 4 rows and 3 columns, what is the degrees of freedom (df) for a chi-square test of homogeneity?

20 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A researcher is conducting a survey on the relationship between college major and career satisfaction. They collected data from 150 students and organized it into a two-way table. If the number of rows in the table represents 3 different college majors and the number of columns represents 5 different levels of career satisfaction, what is the df for a chi-square test of independence?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Constructing a confidence interval

A sample of $n = 121$ has mean $\bar{x} = 393$. A claimed population mean is $\mu_0 = 409$, with sample standard deviation $s = 62$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 2.84

b. -2.34

c. -3.34

d. -2.84

22 [Foundational] • Target: Margin of error and sample size

A sample of $n = 25$ has mean $\bar{x} = 513$. A claimed population mean is $\mu_0 = 495$, with sample standard deviation $s = 75$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -1.20

b. 0.70

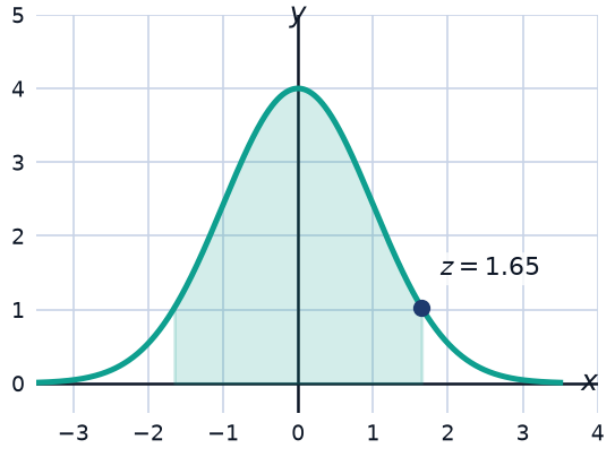
c. 1.70

d. 1.20

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23 [Foundational] • Target: The logic of hypothesis testing

A sample of $n = 25$ has mean $\bar{x} = 469$. A claimed population mean is $\mu_0 = 471$, with sample standard deviation $s = 51$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.



a. -0.20

b. 0.30

c. -0.70

d. 0.20

24 [Foundational] • Target: Four-step hypothesis testing procedure

A sample of $n = 16$ has mean $\bar{x} = 157$. A claimed population mean is $\mu_0 = 183$, with sample standard deviation $s = 34$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -2.56

b. -3.06

c. 3.06

d. -3.56

25 [Foundational] • Target: Type I and Type II errors, and power

A sample of $n = 49$ has mean $\bar{x} = 187$. A claimed population mean is $\mu_0 = 162$, with sample standard deviation $s = 65$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 2.69

b. 3.19

c. -2.69

d. 2.19

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26 [Foundational] • Target: Confidence intervals for a proportion

A sample of $n = 121$ has mean $\bar{x} = 348$. A claimed population mean is $\mu_0 = 346$, with sample standard deviation $s = 28$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 1.29

b. 0.79

c. 0.29

d. -0.79

27 [Foundational] • Target: Determining sample size for a target margin of error

A sample of $n = 64$ has mean $\bar{x} = 241$. A claimed population mean is $\mu_0 = 244$, with sample standard deviation $s = 25$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -0.46

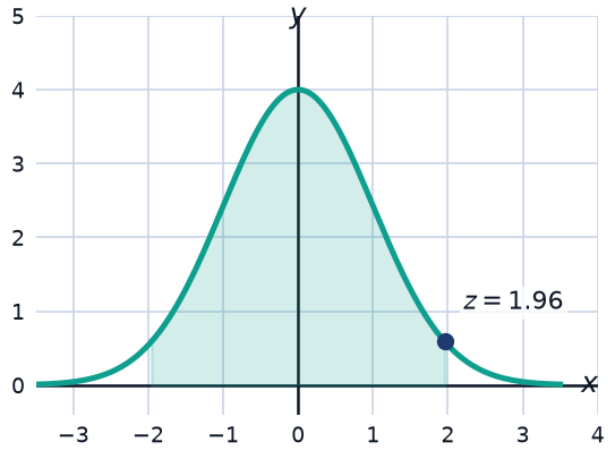
b. -1.46

c. -0.96

d. 0.96

28 [Foundational] • Target: One-sample t-test for a mean

A random sample of $n = 121$ has mean $\bar{x} = 180$ and sample standard deviation $s = 16$. Construct a 99% confidence interval for the population mean, using $z^* \approx 2.576$.



a. (176.25, 181.75)

b. (175.25, 184.75)

c. (178.25, 183.75)

d. (176.25, 183.75)

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Constructing & Interpreting Confidence Intervals

A random sample of $n = 49$ has mean $\bar{x} = 143$ and sample standard deviation $s = 41$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

a. $(130.52, 155.48)$

b. $(133.52, 154.48)$

c. $(131.52, 152.48)$

d. $(131.52, 154.48)$

30 [Intermediate] • Target: Hypothesis Testing Logic & P-values

A random sample of $n = 121$ has mean $\bar{x} = 40$ and sample standard deviation $s = 11$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

a. $(37.04, 42.96)$

b. $(38.04, 41.96)$

c. $(38.04, 39.96)$

d. $(40.04, 41.96)$

31 [Intermediate] • Target: Type I Error, Type II Error, and Power

A random sample of $n = 16$ has mean $\bar{x} = 108$ and sample standard deviation $s = 7$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

a. $(103.57, 112.43)$

b. $(106.57, 111.43)$

c. $(104.57, 111.43)$

d. $(104.57, 109.43)$

32 [Intermediate] • Target: One-Sample and Two-Sample t-Procedures

A random sample of $n = 49$ has mean $\bar{x} = 154$ and sample standard deviation $s = 59$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.

a. $(140.13, 167.87)$

b. $(139.13, 168.87)$

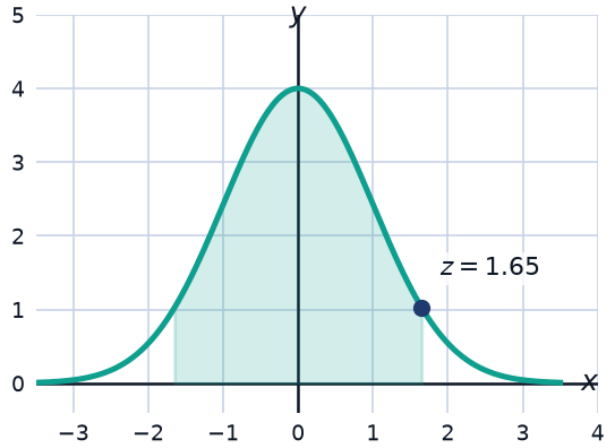
c. $(140.13, 165.87)$

d. $(142.13, 167.87)$

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33 [Intermediate] • Target: Constructing a confidence interval

A random sample of $n = 36$ has mean $\bar{x} = 23$ and sample standard deviation $s = 51$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.



a. $(9.02, 34.98)$

b. $(9.02, 36.98)$

c. $(8.02, 37.98)$

d. $(11.02, 36.98)$

34 [Intermediate] • Target: Margin of error and sample size

A random sample of $n = 16$ has mean $\bar{x} = 22$ and sample standard deviation $s = 56$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

a. $(-3.44, 49.44)$

b. $(-5.44, 49.44)$

c. $(-5.44, 47.44)$

d. $(-6.44, 50.44)$

35 [Intermediate] • Target: The logic of hypothesis testing

A poll of 150 randomly selected voters finds that 37 support a candidate. Construct a 99% confidence interval for the true population proportion, using $z^* \approx 2.576$.

a. $(0.156, 0.307)$

b. $(0.136, 0.357)$

c. $(0.186, 0.337)$

d. $(0.156, 0.337)$

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36 [Intermediate] • Target: Four-step hypothesis testing procedure

A poll of 300 randomly selected voters finds that 143 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.

a. $(0.450, 0.533)$

b. $(0.420, 0.533)$

c. $(0.400, 0.553)$

d. $(0.420, 0.503)$

37 [Intermediate] • Target: Type I and Type II errors, and power

A researcher wants a margin of error of at most 0.02 for a 90% confidence interval for a population proportion, using a planning value of $p = 0.70$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1-p)$ with $z^* = 1.645$.

a. 1421

b. 1420

c. 1371

d. 1471

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Confidence intervals for a proportion • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a planning value of $p = 0.40$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1-p)$ with $z^* = 1.96$.

a. 975

b. 1024

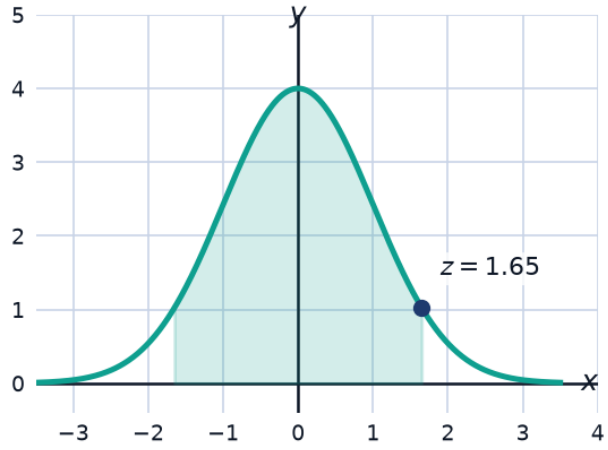
c. 1075

d. 1025

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39 [Challenging] • Target: Determining sample size for a target margin of error • Cross-Lesson: 3.03 The Normal Distribution

A hypothesis test about a nutritionist's claimed mean calorie count produces a p-value of 0.112. Using a significance level of $\alpha = 0.01$, state the correct conclusion.



a. The test is inconclusive and must be repeated

b. Fail to reject H_0; not enough evidence that the true mean calorie count differs from the claim

c. Reject H_0; the true mean calorie count differs from the claim

d. Accept H_0 as true

40 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A hypothesis test about a claimed mean reaction time produces a p-value of 0.034. Using a significance level of $\alpha = 0.05$, state the correct conclusion.

a. The test is inconclusive and must be repeated

b. Accept H_0 as true

c. Fail to reject H_0; not enough evidence that the true mean reaction time differs from the claim

d. Reject H_0; the true mean reaction time differs from the claim

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41 [Challenging] • Target: Constructing & Interpreting Confidence Intervals • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

What is the correct interpretation of a 95% confidence interval for a population mean $(42.1, 48.7)$?

a. We are 95% confident that the true population mean falls between 42.1 and 48.7

b. There is a 95% probability that the true mean is between 42.1 and 48.7

c. 95% of all individual values in the population lie between 42.1 and 48.7

d. 95% of sample means will equal the population mean

42 [Challenging] • Target: Constructing & Interpreting Confidence Intervals • Cross-Lesson: 1.01 Measures of Center & Spread

How can you decrease the width of a confidence interval?

a. Increase the sample size or decrease the confidence level

b. Increase the confidence level or increase the sample size

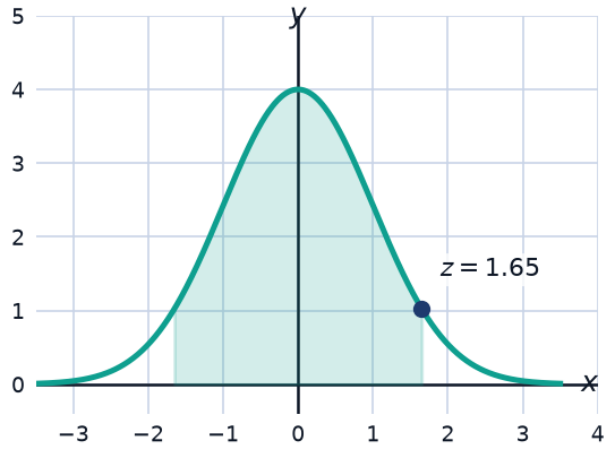
c. Decrease the sample size or increase the confidence level

d. Increase the standard deviation of the data

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43 [Challenging] • Target: Hypothesis Testing Logic & P-values • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

Why do statisticians state 'fail to reject H_0' instead of 'accept H_0'?



a. Failing to find sufficient evidence against H_0 does not prove that H_0 is true

b. Because H_0 is always false in practice

c. To avoid computing the test statistic

d. Because the P-value is always approximate

44 [Challenging] • Target: Type I Error, Type II Error, and Power • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

What is a Type II error?

a. Failing to reject the null hypothesis when the null hypothesis is actually false

b. Rejecting the null hypothesis when it is true

c. Selecting a biased sample

d. Misinterpreting the P-value

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45 [Challenging] • Target: One-Sample and Two-Sample t-Procedures • Cross-Lesson: 2.01 Sampling Methods & Bias

When estimating a population mean μ with unknown σ, which distribution and test statistic are used?

a. t-distribution with $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ and $df = n - 1$

b. Standard normal distribution with $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

c. Chi-square distribution with $df = n - 1$

d. F-distribution with $df = (n - 1, n - 2)$

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